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Integrated Metabolomic and Transcriptomic Analysis Reveals the Basis for the Difference in Flavonoid Accumulation in Six Medicinal Tissues of Lotus (*Nelumbo nucifera* Gaertn.)

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Abstract: (1) Background: Lotus (*Nelumbo nucifera* Gaertn.) is an aquatic plant with a long history and ornamental, edible, medicinal, and commercial value. All parts of the lotus are important "homologous and different effects" medicinal herbs as distinct parts of the same plant with different medicinal effects. (2) Methods: Six representative medicinal parts of lotus were selected, i.e., the lotus flower (Nelumbinis Flos, Flo), lotus petiole (Nelumbinis Petiolus, Pet), lotus seedpod (Nelumbinis Receptaculum, Rec), lotus stamen (Nelumbinis Stamen, Sta), lotus seed (Nelumbinis Semen, Sem), and lotus plumule (Nelumbinis Plumula, Plu), and analyzed via UPLC-MS/MS for metabolite profiling and RNA sequencing for gene expression. Weighted gene co-expression network analysis identified several transcription factor families related to flavonoid synthesis. (3) Results: Distinct metabolite accumulation and gene expression patterns were observed, particularly in the flavonoid synthesis pathway. Key findings included important metabolic profiles in stamens, high flavonoid content in light-exposed tissues, and functional differentiation in seeds and plumules. The discovered transcription factor families are essential for flavonoid biosynthesis and contribute to molecular breeding and resource utilization. (4) Conclusion: This study provides an important theoretical basis for molecular breeding of lotus, quality control of medicinal materials, and the rational use of resources obtained from different medicinal parts of lotus.

Keywords: lotus; metabolomics; transcriptomics; flavonoid biosynthesis; homologous and different effects; WGCNA

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1. Introduction

Lotus (*Nelumbo nucifera* Gaertn.) is a perennial aquatic herb of the genus *lotus* in the family *Lotusaceae*, with ornamental, edible, medicinal, and economic value. Lotus has been cultivated for at least 7000 years in China, in nearly all its provinces. It has also been used as a functional food and herbal medicine in Asia for 2000 years [1]. Lotus seed, lotus plumule, lotus seedpod, lotus leaf, lotus petiole, lotus stamen, lotus root, and lotus flower are all derived from the lotus; however, they are important "homologous and different effects" medicinal herbs as distinct parts of the same plant with different medicinal effects [2].

Among them, lotus leaf, lotus seed, lotus plumule, lotus root, lotus stamen, and lotus seedpod are included in the Chinese Pharmacopoeia. The lotus seed and lotus plumule can effectively treat insomnia and spermatorrhea [3–8]. In addition, lotus seed nourishes the heart, calms the mind, strengthens the spleen, and treats diarrhea; on the other hand, lotus plumule also relaxes fire and treats delirium [6,8–12]. Lotus leaf, lotus seedpod, and lotus root can stop bleeding [7–9,12–15]. The lotus leaf also strengthens the spleen and

treats diarrhea; the lotus seedpod treats postpartum stasis, and the lotus root removes blood stasis [7–9,15]. Researchers have demonstrated that the lotus stamen can enhance kidney and astringent essence and treat frequent urination [8,9,15]. Two herbs, lotus flower and lotus petiole, are not included in the Chinese Pharmacopoeia. Nonetheless, they are included in the Chinese medicinal tablet's concoction specifications of Jiangsu, Shanghai, Hubei, and other regions; they are also used as Chinese local medicines. Lotus flower and lotus petiole can eradicate summer heat. Lotus flowers can also stop bleeding, as well as exert anti-diabetic, anti-obesity effects and other properties [16]. Lotus petiole can also improve chest tightness and treat diseases including dysentery and gonorrhea [17].

Research on the isolation of chemical constituents from different lotus tissues has revealed that the varying efficacy of the lotus's medicinal parts is due to differences in their chemical composition, particularly in regard to metabolites. These chemical components primarily include flavonoids, alkaloids, polyphenols, triterpenoids, proteins, polysaccharides, starch, steroids, and other components [18,19]. Lotus leaf, lotus petiole, lotus seed, and lotus plumule are dominated by alkaloidal components [2,18], with major pharmacological effects including antioxidant activity and anti-tumor, antibacterial, anti-inflammatory, lipid-lowering, and hypoglycemic effects [20]. Flavonoids are primarily found in light-exposed tissues of the lotus, including the leaf, flower, and seedpod [18]. These compounds demonstrate diverse pharmacological activities, such as antioxidant, antiviral, antibacterial, anti-inflammatory, and analgesic effects. Additionally, they show potential in managing conditions like Alzheimer's disease, hypertension, and diabetes mellitus [7,16,18,21,22].

While lotus has been extensively studied for its phytochemical constituents and traditional medicinal applications, the molecular mechanisms underlying the pharmacological activity differences among various tissues remain unclear. This knowledge gap substantially limits the in-depth development and quality control of lotus resources. Omics technologies, including metabolomics, transcriptomics, and proteomics, offer comprehensive insights into the dynamic changes and functional mechanisms of biological systems at different molecular levels, serving as crucial tools to address this issue.

Recent advancements in lotus research employing these omics approaches have yielded critical insights. Metabolomic profiling revealed distinct developmental transitions in lotus maturation; the transition of lotus seed from metabolically highly active tissue to the preparation of storage tissue occurs at 15 days post pollination [23], while the plumule exhibits a metabolic transition to significant secondary metabolite accumulation by day 18 [24]. Comparative analyses across cultivars established strong correlations between antioxidant capacity differentials and specific metabolite profiles [25]. The molecular basis of phenotypic diversity has been partially elucidated through anthocyanin accumulation patterns governing floral pigmentation [26] and MITE-associated variants influencing lotus traits [27]. Genomic investigations have further identified genomic bases for lotus flower color, carpel thermogenesis, and domestication [28], as well as candidate genes for stress resistance in lotus [29]. Systematic multi-omics investigations covering lotus tissues have progressively mapped the regulatory networks governing lotus metabolism [30–34].

However, multi-omics research on the lotus genus is still in its nascent stages, and is lacking a systematic analysis of tissue metabolite distribution and key metabolic pathways. In particular, the genetic basis of tissue-specific metabolic differences remains to be elucidated. Therefore, it is imperative to investigate the tissue-specific distribution of metabolites and gene expression in lotus plant to determine the mechanisms underlying the "homologous and different effects" phenomenon of medicinal parts of plants. This will also provide a theoretical foundation for the comprehensive utilization of lotus resources and the establishment of quality standards for certain lotus species used in medicinal applications.

2. Materials and Methods

2.1. Materials

The material used in this study was harvested from Nanchang, Jiangxi Province in June 2024, and was identified as lotus (*Nelumbo nucifera* Gaertn.) by Dr. Yihao Jiang of Nanchang University. Six representative medicinal parts of the lotus plant were selected, i.e., the lotus flower (Nelumbinis Flos, Flo), lotus petiole (Nelumbinis Petiolus, Pet), lotus seedpod (Nelumbinis Receptaculum, Rec), lotus stamen (Nelumbinis Stamen, Sta), lotus seed (Nelumbinis Semen, Sem), and lotus plumule (Nelumbinis Plumula, Plu), with three biological replicates for each tissue sample. After harvesting, the samples were washed and wiped clean, immediately frozen in liquid nitrogen, and placed in a −80 °C cryogenic refrigerator for spare parts.

2.2. Metabolite Extraction Methods

Non-targeted analysis was performed for the identification of metabolites. The extraction methods used in this study were described previously by Vasilev et al. (2016) [35]. The sample was weighed into a 2 mL centrifuge tube and vortexed with 600 µL of methanol containing 2-chloro-L-phenylalanine (4 ppm) for 30 s. Steel beads were added to the sample and placed into a tissue grinder (MB-96) at 55 Hz for 60 s. The sample was sonicated for 15 min at room temperature and centrifuged at 12,000 rpm at 4 °C for 10 min. The supernatant was filtered using a 0.22 µm membrane, and the filtrate was transferred to the detection vial for LC-MS analysis.

2.3. UPLC-MS/MS Conditions

Metabolome analysis was conducted on a Thermo Vanquish (Thermo Fisher Scientific, Waltham, MA, USA) ultra-high-performance liquid chromatography (UPLC) system using an ACQUITY UPLC® HSS T3 (2.1 × 100 mm, 1.8 µm) (Waters, MiRecord, Milford, MA, USA) column at a flow rate of 0.3 mL/min and an injection volume of 2 µL at a column temperature of 40 °C. In the positive ion mode, the mobile phases were 0.1% formic acid in acetonitrile (B2) and 0.1% formic acid in water (A2). The gradient elution program was as follows: 0~1 min, 8% B2; 1~8 min, 8%~98% B2; 8~10 min, 98% B2; 10~10.1 min, 98%~8% B2; 10.1~12 min, 8% B2. In the negative ion mode, the mobile phases were acetonitrile (B3) and 5 mM ammonium formate in water (A3), and the gradient elution procedure was as follows: 0~1 min, 8% B3; 1~8 min, 8%~98% B3; 8~10 min, 98% B3; 10~10.1 min, 98%~8% B3; 10.1~12 min, 8% B3.

Mass spectrometry analysis was performed using a Thermo Q Exactive Focus instrument (Thermo Fisher Scientific, USA) equipped with an electrospray ionization (ESI) source operating in both positive and negative ion modes. Instrument parameters were set as follows: spray voltages at 3.50 kV (positive) and −2.50 kV (negative), sheath gas flow 40 arb, auxiliary gas 10 arb, capillary temperature 325 °C. Full scan acquisition was conducted with 70,000 resolution over m/z 100–1000, followed by data-dependent MS/MS acquisition (top 3 precursors) using higher-energy collisional dissociation (HCD) at 30 eV with 17,500 resolution. Dynamic exclusion was applied to minimize redundant fragmentation [36].

2.4. Metabolite Profiling

The metabolomics data were preprocessed using support vector regression (SVR) calibration based on quality control (QC) samples to eliminate systematic errors. Compounds with relative standard deviation (RSD) values greater than 30% in the QC samples were filtered out during the quality control and quality assurance process before subsequent

data analysis. This data preprocessing approach assesses data quality and corrects for instrumental bias and further enhances data quality through RSD-based filtering.

A spectrum library of plant substances including primary and secondary plant metabolites was constructed with reference to standards derived from several public databases including HMDB, MassBank, KNApSAcK, ReSpect, LipidMaps, and KEGG. Metabolite annotation was achieved through orthogonal matching of accurate mass, retention time, and MS/MS spectral similarity. Differentially abundant metabolites (DAMs) were identified using multivariate statistics (VIP > 1) and univariate testing (p-value < 0.05) [37]. Functional pathway enrichment and topological analyses were conducted for the identified differential metabolites using the MetaboAnalyst 5.0 software package. The pathways identified in different enrichment analyses were subjected to further screening using the KEGG Mapper visualization tool.

2.5. RNA-Seq

Libraries were created using the NEBNext Ultra II RNA Library Prep Kit for Illumina. Library quality was checked using an Agilent 2100 Bioanalyzer (Agilent, 2100, Santa Clara, CA, USA) and Agilent High Sensitivity DNA Kit (Agilent, 5067-4626). Total library concentration was detected using Pico green (Quantifluor-ST fluorometer, Promega, Madisson, WI, USA, E6090; Quant-iT PicoGreen dsDNA Assay Kit, Invitrogen, P7589, Waltham, MA, USA) and validated library concentration was quantified by QPCR <Thermo Scientific, Waltham, MA, USA, StepOnePlus Real-Time PCR Systems>. The mixed libraries were diluted step-wise and quantified before sequencing in PE150 mode on an Illumina (San Diego, CA, USA) sequencer.

The filtered high-quality sequences (Clean Data) were aligned to the reference genome of the species. Moreover, the Read Count value of each gene was counted using HT-Seq (0.9.1) as the raw expression of the gene. Expression was normalized (Normalization) using FPKM to assess gene expression levels. Differential expression analysis between two comparison combinations was performed using DESeq (1.20.0). Differentially expressed genes (DEGs) were screened under the conditions of expression difference ploidy $|\log2(\text{FoldChange})| > 1$, and significance p-value < 0.05.

2.6. Differential Gene Analysis

KEGG pathway enrichment analysis of differential genes was carried out using topGO with clusterProfiler (3.4.4). p-values were computed using hypergeometric distribution methods to identify pathways significantly enriched for differential genes, thereby determining the major biological functions of differential genes.

Further, protein interactions were analyzed based on the STRING database (https://string-db.org (accessed on 8 July 2024)), hence revealing the relationships between target genes and interrelationships between proteins of the species.

2.7. Weighted Gene Co-Expression Network Analysis

Gene expression levels were obtained by RNA-seq, and the gene co-expression network was constructed using the WGCNA software (version 1.72-1) to achieve efficient mining of sequencing data, classify the data, and select the major effector genes. The co-expression similarity coefficients between genes were calculated using the TOMSimilarity module; a power value of 20 was selected using the pickSoftThreshold function of this software package when the correlation coefficients stabilized. We selected the modules with correlation coefficients of >0.8 to the grouped features. Cytoscape 3.10.2 was utilized to draw the module gene network diagrams.

3. Results

3.1. Differential Analysis of Secondary Metabolism in Lotus Tissues

The total ion chromatogram (TIC) of six tissues of the lotus plant in positive and negative ion modes are shown in Figure 1A,B, respectively. After screening differential metabolites from the confidently identified metabolite list, 8638 and 8125 differential metabolites were identified in positive and negative ion modes, respectively. Principal component analysis (Figure 1C,D) of metabolites in positive and negative ion modes showed differences in metabolites between groups of different tissues; differences within groups were not apparent, which could explain the unique metabolic characteristics of different tissues to a certain extent. The Nelumbinis Stamen (Sta) was clearly distinguished from other tissues with unique metabolic features. The Nelumbinis Semen (Sem) and Nelumbinis Plumula (Plu) had similar metabolic characteristics. Notably, the Nelumbinis Petiolus (Pet), Nelumbinis Receptaculum (Rec), and Nelumbinis Flos (Flo) are the light-seeing tissues of the lotus growing on the water surface. Flo, Rec, and Sem are the flower, the fruit, and the seed of the lotus. The lotus pollen comes from Sta, and after successful pollination, the pistil develops into Rec; inside Rec, the Sem develops, and the embryonic root, Plu, develops in Sem. Therefore, we set up the following comparison combinations: Flo vs. Rec, Flo vs. Pet, Rec vs. Pet, Flo vs. Sem, Rec vs. Sem, Sta vs. Rec, Rec vs. Plu, Sem vs. Plu, and Sta vs. Sem.

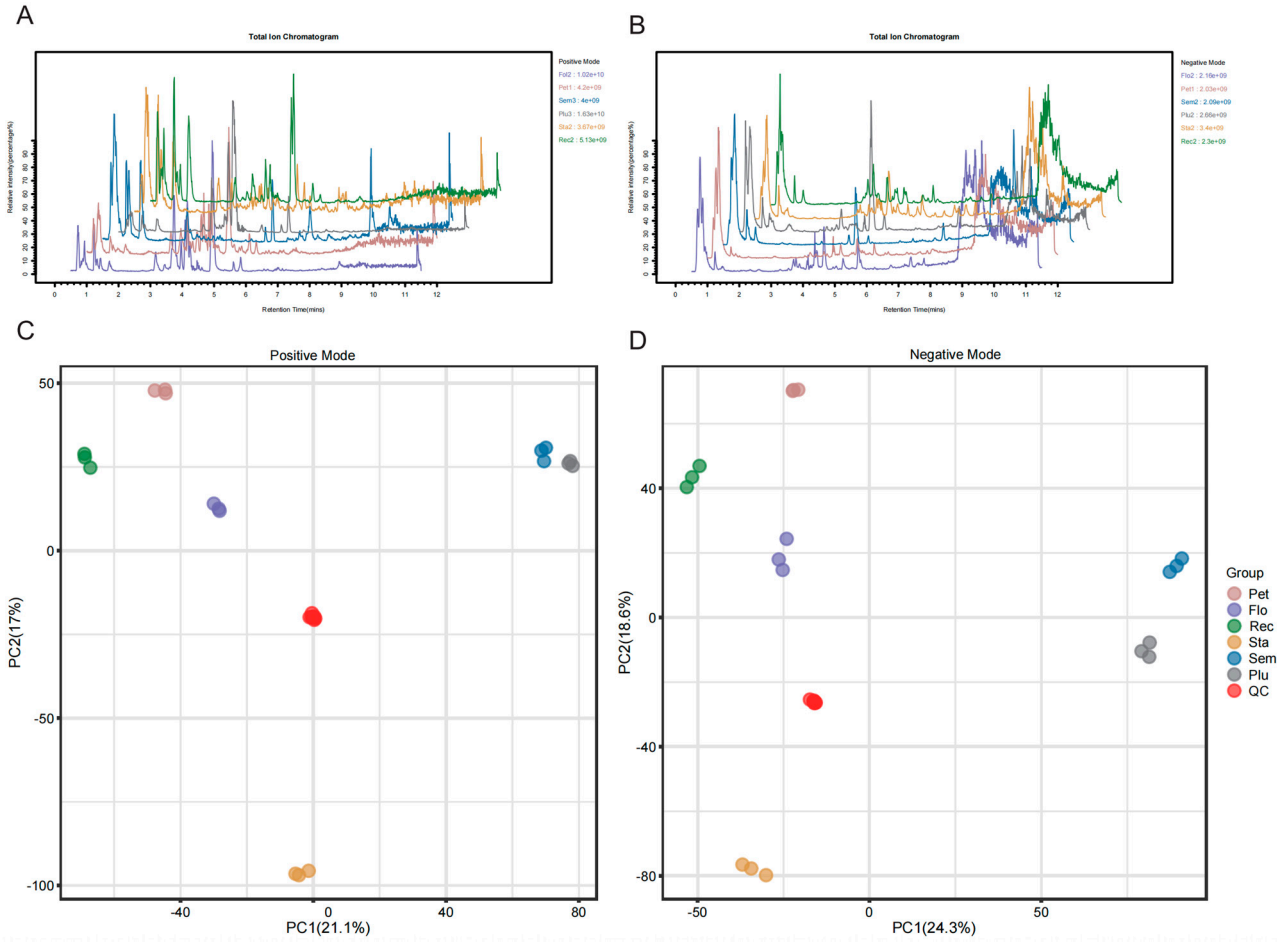


Figure 1. *Cont.*

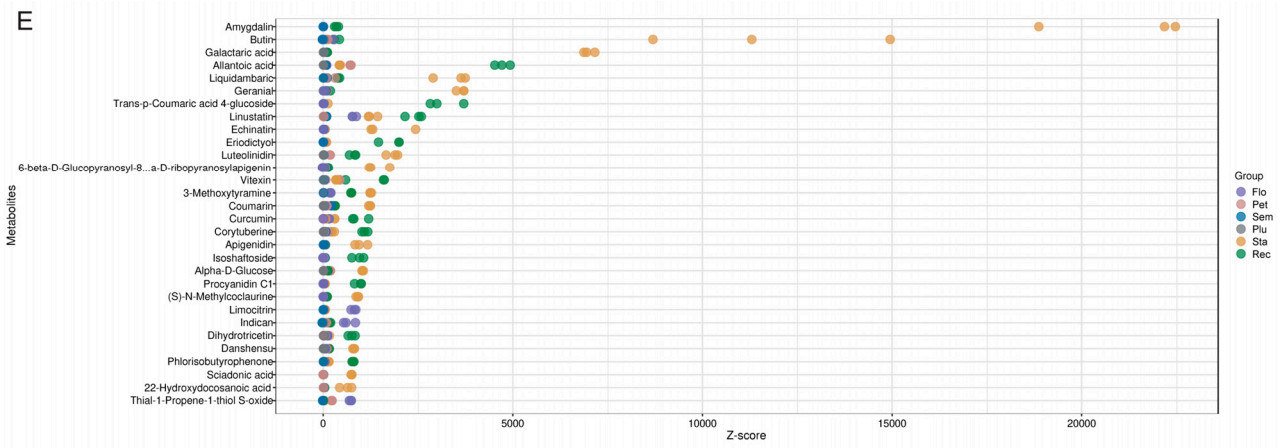


Figure 1. The TIC chromatograms, PCA score plot, and Z-score graph of six different tissues in lotus (Pet, Flo, Rec, Sta, Sem, and Plu). (**A**,**B**) The TIC chromatograms in positive and negative modes. (**C**,**D**) The PCA score plots in positive and negative modes. (**E**) The Z-score graph of different tissues. (The top 30 differential metabolites with high scores were presented).

The Z-score plot can visualize the overall trend and degree of difference in the quantitative values of metabolites among different groupings and samples. Figure 1E shows the top 30 differential metabolites, which exhibited the most significant degree of variation among the six tissues of the lotus. All the metabolites with a greater degree of variation were significantly enriched in Sta. Based on the results of the principal component analysis, the Z-score plot displayed a unique metabolic profile of Sta. Amygdalin, which has the highest variation among the six tissues and the highest relative content in Sta, is usually broadly distributed in Rosaceae [38], and more commonly found in almond seed kernels as well as apricot pollen [39,40]. Pharmacological effects include cough suppressant, anticancer, anti-inflammatory, analgesic, and hypoglycemic effects [38,41]. In addition, metabolite contents including butin, galactaric acid, liquidambaric, geranial, and echinatin in Sta were significantly higher than that in the other five tissues.

MS/MS secondary analysis identified 1094 secondary metabolites in six tissues of the lotus plant (Figure 2A, Table S1), including 122 flavonoids, 95 prenol lipids, 92 organooxygen compounds, 86 fatty acyls, 57 carboxylic acids and derivatives, 54 benzene and substituted derivatives, 50 phenols, 40 terpenoids, and 32 alkaloids and other compounds.

A total of 92 differential metabolites were common to different comparative combinations in different tissues of lotus, including 16 flavonoids, 7 phenols, 6 terpenoids, 6 organooxygen compounds, 4 alkaloids, 4 benzene and substituted derivatives, and 4 prenol lipids and other compounds. Among them, flavonoids had the largest percentage, showing that they had the greatest effect on the differentiation of different tissues of the lotus (Figure 2B).

To investigate the accumulation pattern of secondary metabolites in different tissues of the lotus, K-means cluster analysis was performed on 1094 secondary metabolites identified in six tissues of the lotus plant (Figure 2C, Table S2). The findings revealed that secondary metabolites accumulated in different tissues with different patterns. The secondary metabolites were classified into nine different clusters based on the distribution trend of their contents in different tissues. Clusters 1 and 7 contained 125 and 116 secondary metabolites, most of which were organooxygen compounds, with a trend of specific accumulation in Pet and Sem. Cluster 3, Cluster 5, and Cluster 9 had 109, 152, and 160 secondary metabolites, respectively, which were majorly flavonoids, and specifically accumulated in Flo, Plu, and Rec, respectively. Cluster 8 had a total of 252 secondary metabolites, specifically accumulating in Sta, most of them being prenol lipids.

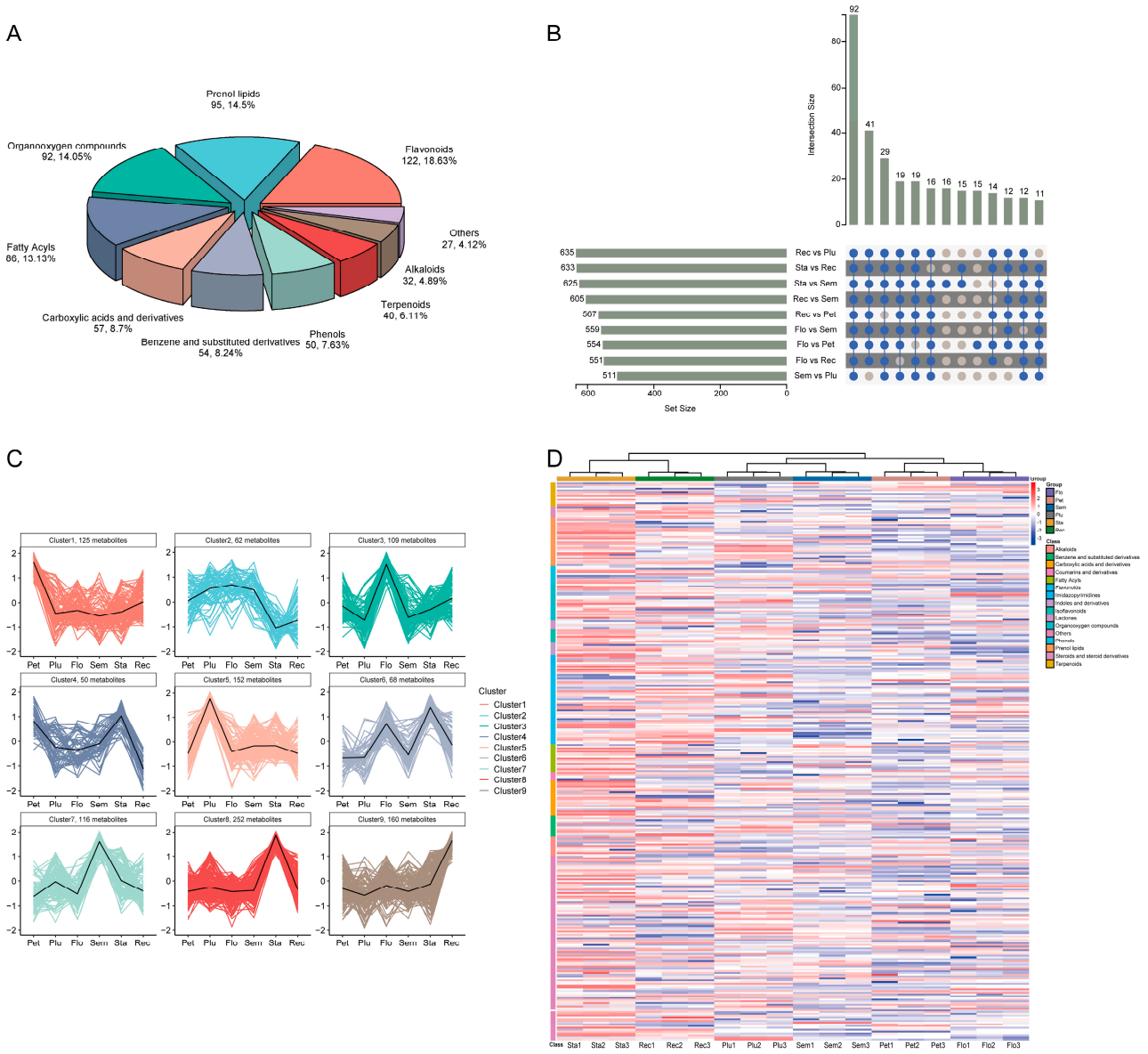


Figure 2. Metabolomic profiling of lotus. (**A**) Pie chart of secondary metabolites classification. (**B**) Upset plot of DAMs shared among the comparison of Flo vs. Rec, Flo vs. Pet, Rec vs. Pet, Flo vs. Sem, Rec vs. Sem, Sta vs. Rec, Sta vs. Sem, Rec vs. Plu, and Sem vs. Plu. (**C**) K-means clustering grouped the metabolites into 9 clusters. (**D**) Clustering heatmap of the secondary metabolites detected in 6 lotus tissues.

The results of the cluster heat map (Figure 2D) showed that flavonoids, organooxygen compounds, fatty acyls, carboxylic acids and derivatives, benzene and substituted derivatives, phenylpropanoids, and indoles and derivatives were significantly enriched in Sta compared to other tissues. Additionally, the constituents enriched in Rec were primarily lactones, and those in Plu were mainly phenols, terpenoids, steroids and steroid derivatives, and imidazopyrimidines. Prenol lipids were the most abundant component in Sem. The most abundant metabolites in Flo were alkaloids, isoflavones, coumarins, and derivatives.

3.2. Transcriptome Analysis of Lotus Tissues

Transcriptome sequencing was performed for the same samples to explore the molecular basis responsible for the flavonoid differences in different tissues. Consequently, we obtained 78,966,7166 Clean Reads, with a Q30 above 95% (Table 1). The obtained Unigenes

were annotated to 12 major public databases including GO (75.43%), eggNOG (84%), KEGG (40.39%), SwissprotName (80.51%), MF (64.63%), NR (100%), eggNOG_Category (84%), Swissprot (83.5%), CC (40.14%), Pathway (21.7%), BP (52.9%), and Entrez_geneID (100%).

Table 1. Quality information of transcriptome sequencing of lotus.

Sample	Reads No.	Clean Reads No.	Clean Data (bp)	N (%)	Q30 (%)
Pet1	44796076	44195988	6663704056	0.004196	96.85
Pet2	45964130	45336206	6825848171	0.004153	96.8
Pet3	50239706	49472494	7454090951	0.007693	96.06
Plu1	41582154	40895430	6151047640	0.007237	95.59
Plu2	46805894	46177374	6952022595	0.007229	96.18
Plu3	54998098	54188514	8134266098	0.007403	95.91
Flo1	40858150	40235688	6066986452	0.006309	96.87
Flo2	41037134	40556292	6118355186	0.006397	97.52
Flo3	45648060	45110000	6801259085	0.006159	97.28
Sem1	44918700	44284232	6680348123	0.006197	97
Sem2	43378918	42815524	6456737375	0.006276	97.12
Sem3	40897244	40409936	6090914512	0.006272	97.33
Sta1	38924410	38349966	5780282858	0.006001	96.9
Sta2	43999936	43409676	6547853705	0.006264	97.03
Sta3	39146748	38586882	5817568921	0.006328	96.9
Rec1	39690262	39175626	5904032544	0.006293	97.07
Rec2	45727084	45109178	6799917596	0.006294	97.03
Rec3	52071532	51358160	7742502649	0.006194	96.98
Total	800684236	789667166	1.18988×10^{11}	-	-

Based on the principal component analysis of the transcriptome (Figure 3A), it is evident that the clustering results between the different tissues were highly consistent with the results of the principal component analysis of the metabolome. Sta remained clustered in a separate group with a unique expression profile. Pet, Rec, and Flo also had similar expression features. In contrast, Sem and Plu were no longer clustered into one group.

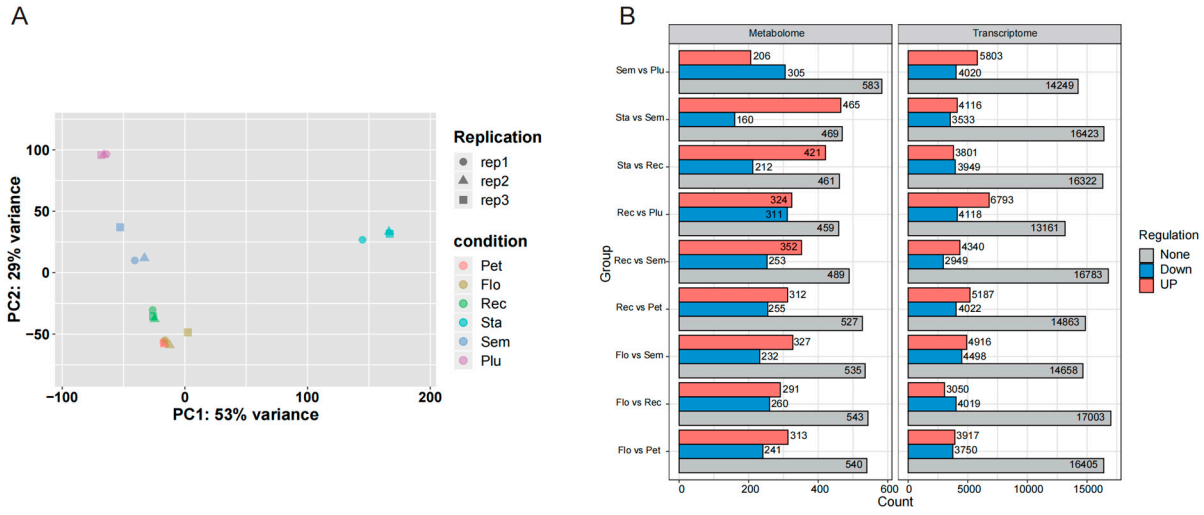


Figure 3. Transcriptomic principal component analysis and histogram of differential molecule numbers in the lotus plant. (**A**) The PCA score plot of 6 lotus tissues. (**B**) Histogram of differential molecules detected by metabolomics and transcriptomics.

3.3. Joint Metabolomic and Transcriptomic Analysis of Lotus Tissues

The joint analysis of metabolism and transcription can unravel the relationship pattern of metabolites and genes and reveal the regulatory mechanism of metabolic pathways in organisms. Based on the number of differential molecules obtained from metabolome and transcriptome assays, the most DAMs were found between Rec and Plu, with 324 DAMs up-regulated and 311 DAMs down-regulated. This was followed by Sta with Rec, with 421 DAMs up-regulated and 212 DAMs down-regulated. The fewest DAMs were found between Sem and Plu, with 206 DAMs up-regulated and 305 DAMs down-regulated. The DEGs between Plu and Rec were the most, with 6793 DEGs up-regulated and 4118 DEGs down-regulated. The next largest number of DEGs was between Plu and Sem, with 5803 DEGs up-regulated and 4020 DEGs down-regulated. Rec and Flo had the least DEGs with 3050 up-regulated and 4019 down-regulated. This shows that the extent of difference in metabolites and genes between the two tissues is not strongly correlated with the distance of growth from each other (Figure 3B).

From the KEGG pathway enrichment results of DAMs and DEGs, the top 20 pathways with the strongest degree of enrichment were selected to draw KEGG pathway bubble maps (Figure 4). In the nine sets of comparison combinations (Flo vs. Rec, Flo vs. Pet, Rec vs. Pet, Flo vs. Sem, Rec vs. Sem, Rec vs. Plu, Sem vs. Plu, Sta vs. Rec, Sta vs. Sem), the flavonoid biosynthesis pathway was enriched in both of the two omics approaches in six of these groups (Flo vs. Rec, Flo vs. Pet, Rec vs. Pet, Flo vs. Sem, Rec vs. Sem, Sta vs. Rec). The pathway was enriched to 25 DAMs and 31 DEGs in Flo vs. Rec, 26 DAMs and 35 DEGs in Flo vs. Pet, 22 DAMs and 37 DEGs in Rec vs. Pet, 25 DAMs and 35 DEGs in Flo vs. Sem, 23 DAMs and 29 DEGs in Rec vs. Sem, and 24 DAMs and 28 DEGs in Sta vs. Rec. KEGG pathway bar graphs were plotted based on the enrichment results (Figure 5), with the red horizontal line indicating a p-value of 0.01 and the blue horizontal line indicating a p-value of 0.05. The flavonoid biosynthesis pathway had a p-value <0.05 in both two omics approaches of the six comparative combinations, suggesting that this pathway is the key regulatory pathway; hence, DAMs and DEGs on this pathway were selected for further analysis.

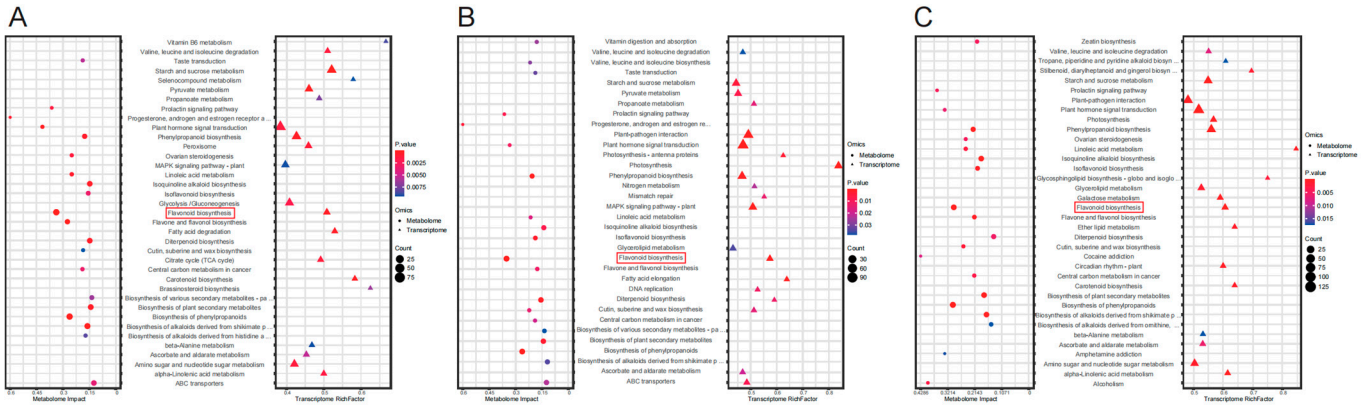


Figure 4. *Cont.*

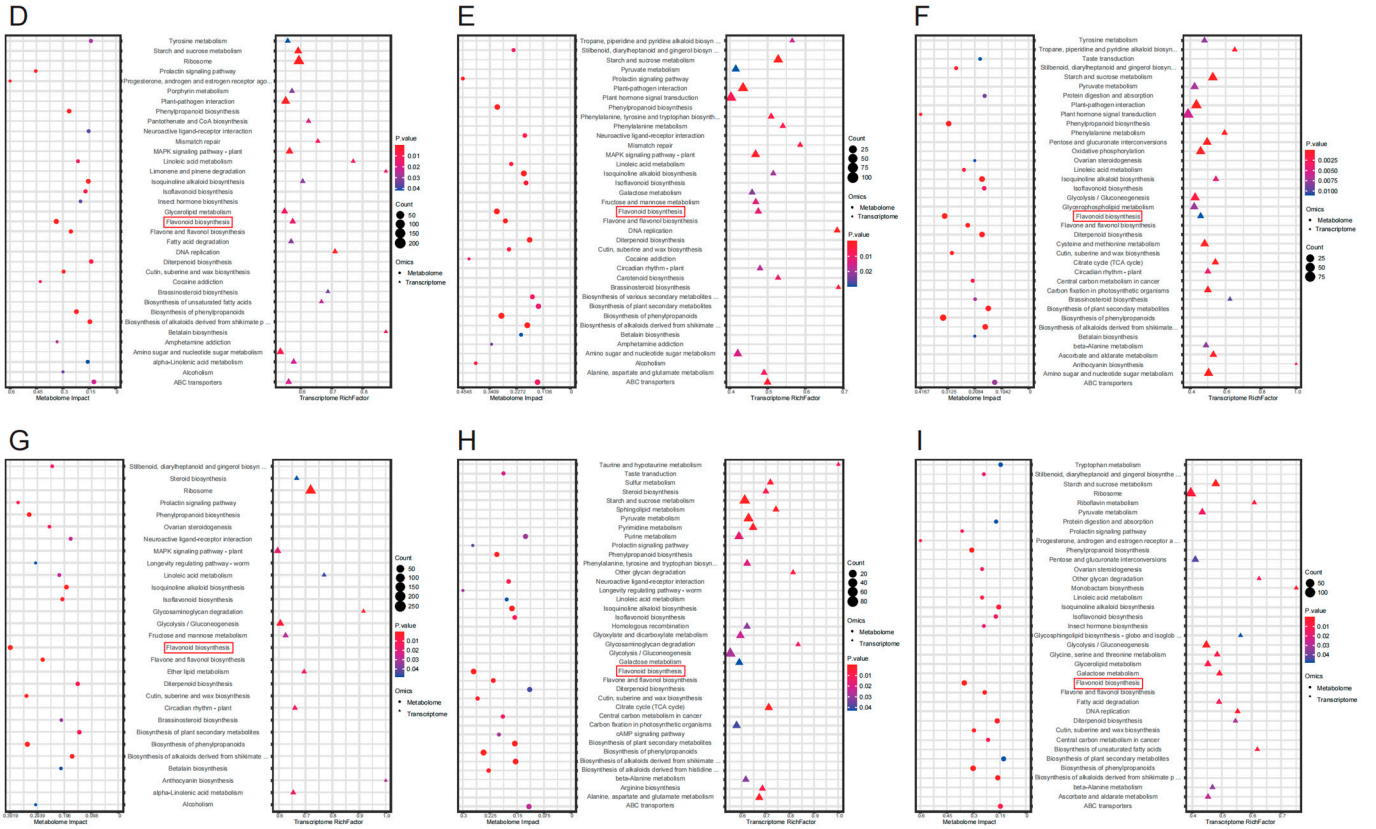


Figure 4. KEGG enrichment bubble map. (**A**) Flo vs. Rec; (**B**) Flo vs. Pet; (**C**) Rec vs. Pet; (**D**) Flo vs. Sem; (**E**) Rec vs. Sem; (**F**) Sta vs. Rec; (**G**) Rec vs. Plu; (**H**) Sem vs. Plu; (**I**) Sta vs. Sem.

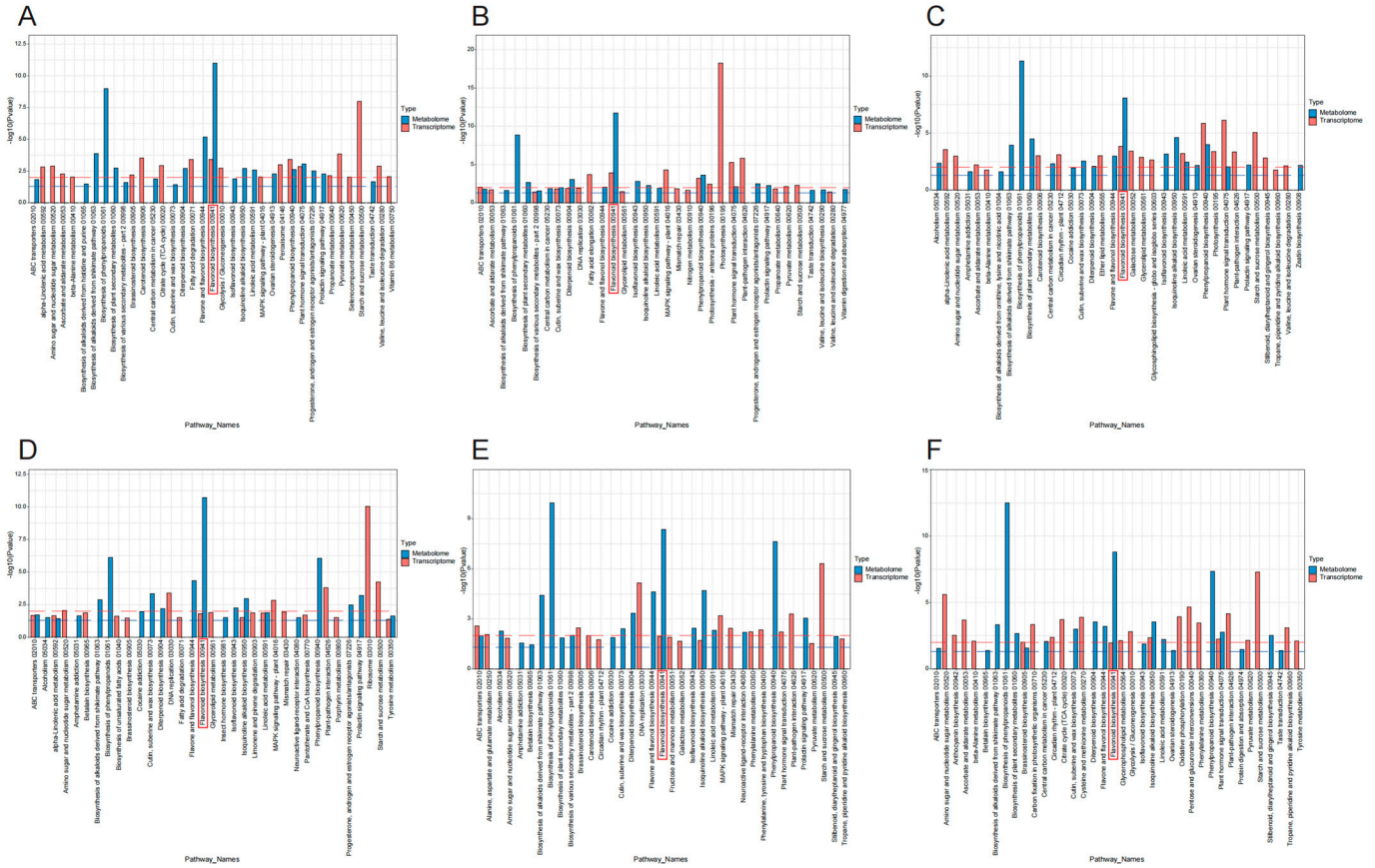


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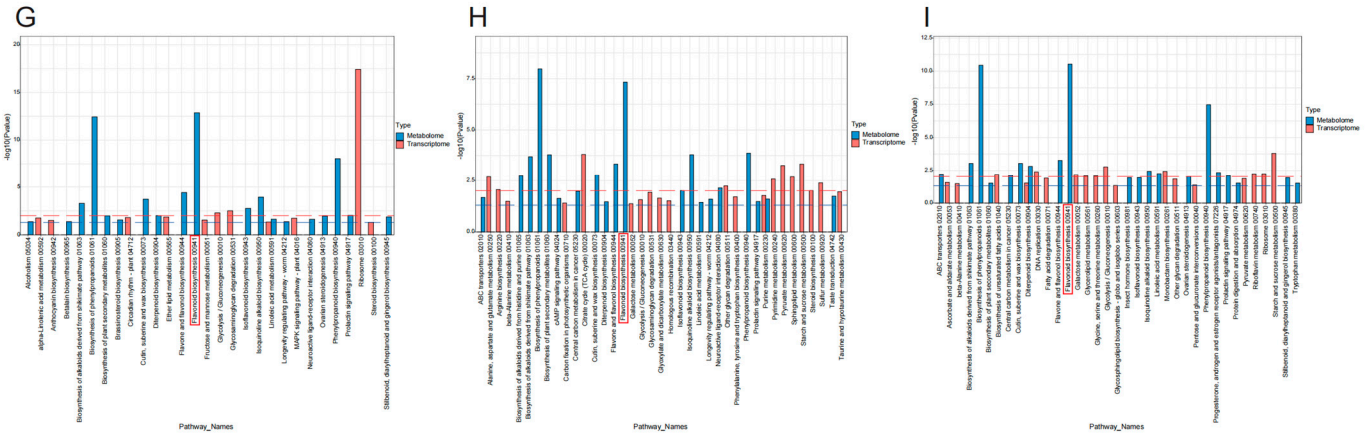


Figure 5. KEGG enrichment histogram. (The red horizontal line indicates that the *p*-value is 0.01, and the blue horizontal line indicates that the *p*-value is 0.05.) (**A**) Flo vs. Rec; (**B**) Flo vs. Pet; (**C**) Rec vs. Pet; (**D**) Flo vs. Sem; (**E**) Rec vs. Sem; (**F**) Sta vs. Rec; (**G**) Rec vs. Plu; (**H**) Sem vs. Plu; (**I**) Sta vs. Sem.

To gain deeper insights into the accumulation mechanism of flavonoid compounds, the levels of differential metabolites and the expression patterns of differential genes within the pathway were visualized using heatmaps. These heatmaps were further integrated with the KEGG pathway to construct comprehensive metabolites and genes heatmaps for the flavonoid biosynthesis pathway (Figure 6).

A total of 34 differential flavonoid components, 64 key-acting genes, and 17 key synthases were annotated in the flavonoid biosynthesis pathway. These included phenylalanine ammonia-lyase (*PAL*), 4-coumarate-CoA ligase (*4CL*), chalcone synthase (*CHS*), trans-cinnamate 4-monooxygenase (*CYP73A*), shikimate O-hydroxycinnamoyltransferase (*HCT*), 5-O-(4-coumaroyl)-D-quinate 3′-monooxygenase (*C3′H*), caffeoyl-CoA O-methyltransferase (*CCoAOMT*), chalcone isomerase (*CHI*), phlorizin synthase (*PGT1*), naringenin 3-dioxygenase (*F3H*), flavonoid 3′-monooxygenase (*F3′H*), flavonoid 3′,5′-hydroxylase (*F3′5′H*), bifunctional dihydro flavonol 4-reductase (*DFR*), flavanol synthase (*FLS*), anthocyanidin synthase (*ANS*), leucoanthocyanidin reductase (*LAR*), and anthocyanidin reductase (*ANR*).

Notably, *CHS* catalyzes the production of various types of chalcone, acting as the first rate-limiting enzyme of the flavonoid biosynthesis pathway. Their expression levels can directly indicate the amount of flavonoids synthesized by the plant body [42,43]. Our findings showed that flavonoid components of Cluster 9 were significantly enriched in Rec, which may be because the *CHS*s were expressed at the highest level in Rec.

CHI catalyzes the production of flavanones from chalcone and is the second rate-limiting enzyme on the flavonoid biosynthesis pathway [44]. The gene heatmap results revealed that *CHI*s were most highly expressed in Pet, leading to the highest flavanone content in this tissue. This included compounds such as pinocembrin, liquiritigenin, and pinobanksin.

F3′H and *F3′5′H* synergistically act on the substrate dihydrokaempferol to generate dihydroquercetin and dihydromyricetin, respectively. The content of dihydroquercetin was significantly higher than that of dihydromyricetin in Rec, Sta, and Sem; this was consistent with the relative expression of the *F3′H*s and the *F3′5′H*s in these three tissues.

It has been reported that *DFR* and *FLS* act on the substrate dihydroflavonols, converting them to colorless anthocyanins and flavonols, respectively [45]. *ANS* competes with *LAR* for the same substrate, colorless anthocyanins, converting colorless anthocyanins to anthocyanidins and proanthocyanidins, which influence the coloring mechanism of the plant and have good antioxidant properties [46]. Although the expression level of *ANS*s was significantly higher in Pet, Rec, and Sem than in other tissues, analysis of the metabolite clustering heatmap showed that Flo had the highest amount of delphinidin, whereas Sta

and Rec contained a higher amount of pelargonidin. This may be attributed to higher enzyme activity of *ANS* in these tissues or to the negative regulatory effect of high *ANR* gene expression on anthocyanins.

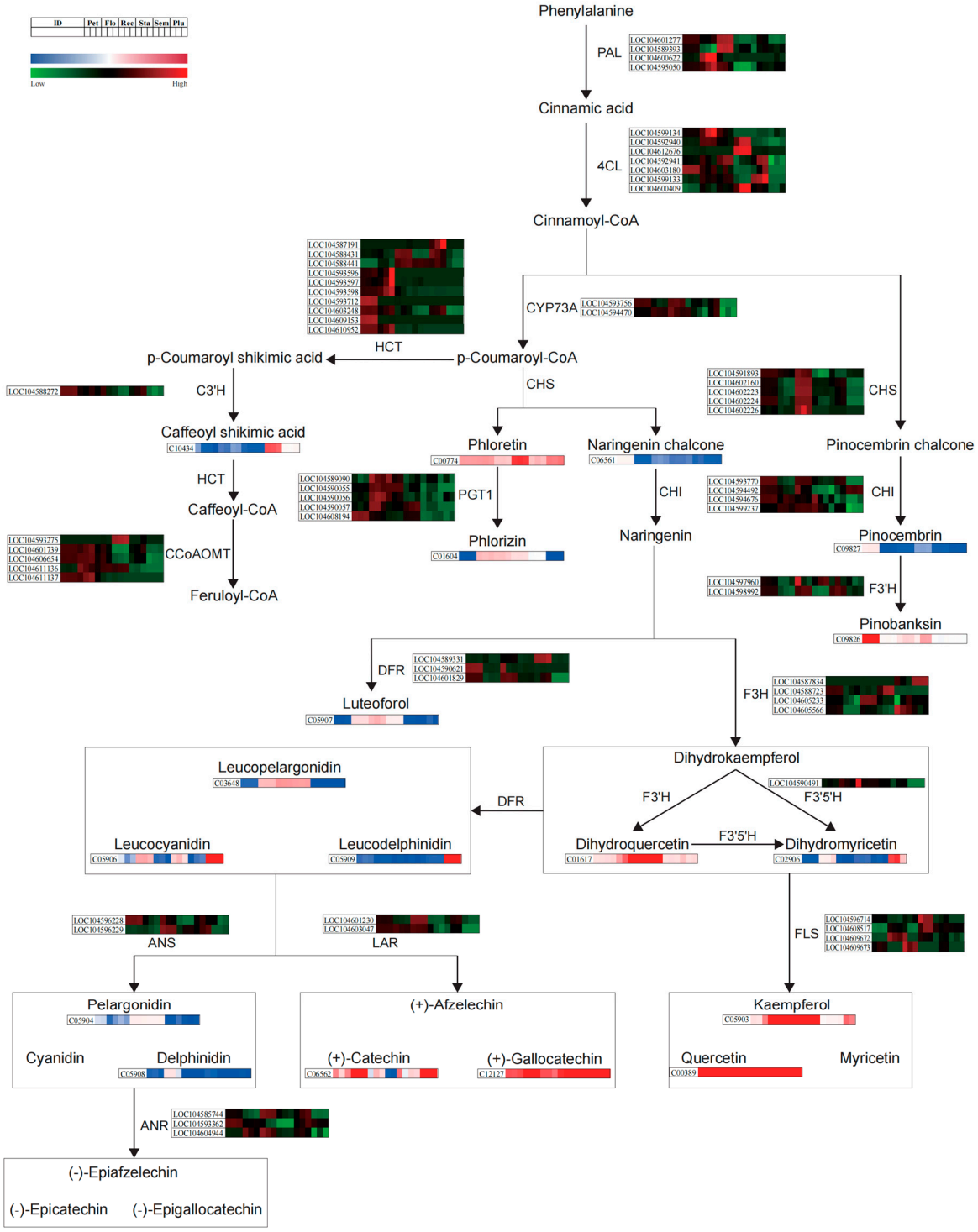


Figure 6. Heatmap of key genes and metabolites expression in flavonoid biosynthesis pathway of 6 lotus tissues.

3.4. WGCNA of DEGs in Lotus Tissues

In further analyses, we screened highly expressed genes from transcriptome data and 10 key flavonoids from differential metabolites of six lotus tissues for WGCNA analysis

to explore the gene expression profile and search for flavonoid-synthesis-related gene modules and co-expressed genes.

Among the 17 co-expressed modules identified, we considered the modules that satisfied the conditions of R > 0.8 and *p*-value < 0.01 as highly related to the corresponding flavonoids. The module–trait heatmap (Figure 7A) showed that the highly expressed genes in Plu were majorly distributed in the blue module, which was highly correlated with (+)-catechin and (+)-gallocatechin. The highly expressed genes in Sta were mainly distributed in the black module and highly correlated with kaempferol. Most of the highly expressed genes in Pet were in the turquoise module and were highly correlated with pinobanksin and naringenin chalcone. In the Rec tissue, genes were predominantly located in the magenta module and were strongly correlated with luteoforol. The red module, which was highly expressed in both Rec and Flo, showed a significant correlation with phlorizin and luteoforol. Meanwhile, the light cyan and yellow modules mainly comprised genes that were highly expressed in Flo and strongly correlated with delphinidin.

Among these modules strongly correlated with flavonoid metabolites, the black module, turquoise module, magenta module, light cyan module, and yellow module contained one, two, six, one, and three structural genes related to the flavonoid biosynthesis pathway, respectively. These five modules were screened as key modules and used to identify the genes in the modules (Table 2, Table S3). The black module contains LOC104608517 (*FLS*) which primarily affects the synthesis of kaempferol. LOC104603248 (*HCT*), and LOC104588272 (*C3′H*) in the turquoise module modulate the synthesis of pinobanksin and naringenin chalcone. The important genes in the magenta module for luteoforol synthesis include LOC104601277 (*PAL*), LOC104589393 (*PAL*), LOC104602160 (*CHS*), LOC104602224 (*CHS*), LOC104605233 (*F3H*), and LOC104585744 (*ANR*). The light cyan module has LOC104611136 (*CCoAOMT*), which is important for delphinidin synthesis. The important genes in the yellow module for delphinidin synthesis include LOC104595050 (*PAL*), LOC104601739 (*CCoAOMT*), and LOC104606654 (*CCoAOMT*).

Among the five key modules listed in Table 2, the structural genes related to flavonoid synthesis, along with their co-expressed genes that exhibit reciprocal relationships within these modules, were visualized in heatmaps (Figure 7B–N). Consequently, the genes included in the different modules had specific expression trends in different tissues of the lotus. These genes have an important effect on the accumulation of flavonoids in different tissues.

To further identify the transcription factors involved in flavonoid synthesis, we analyzed those that were closely associated with the structural genes in the key modules. Subsequently, we constructed an interoperability network diagram using the structural genes (Figure 7O, Table S4), including 38 nodes and 167 edges. The circles represent the transcription factors and squares indicate the structural genes. These 67 transcription factors were from 26 transcription factor families including 10 WRKY, 8 bHLH, 7 ERF, 4 HD-ZIP and TALE, 3 ARF, C2H2, and NAC, and other transcription factor families. The MCC algorithm was used to screen the top 10 nodes with centrality (Table 3). Among the transcription factors analyzed, bHLH demonstrated the highest degree of connectivity, interacting with seven structural genes. ERF was connected to six structural genes, while NF-YC was associated with five. By examining the correlations between structural genes and flavonoid compounds within the modules, we infer that bHLH plays a direct or indirect role in regulating the biosynthesis of kaempferol, pinobanksin, naringenin chalcone, luteoforol, and delphinidin. ERF is involved in the regulation of the synthesis of pinobanksin, naringenin chalcone, luteoforol, and delphinidin. NF-YC is involved in the regulation of pinobanksin, naringenin chalcone, and luteoforol synthesis.

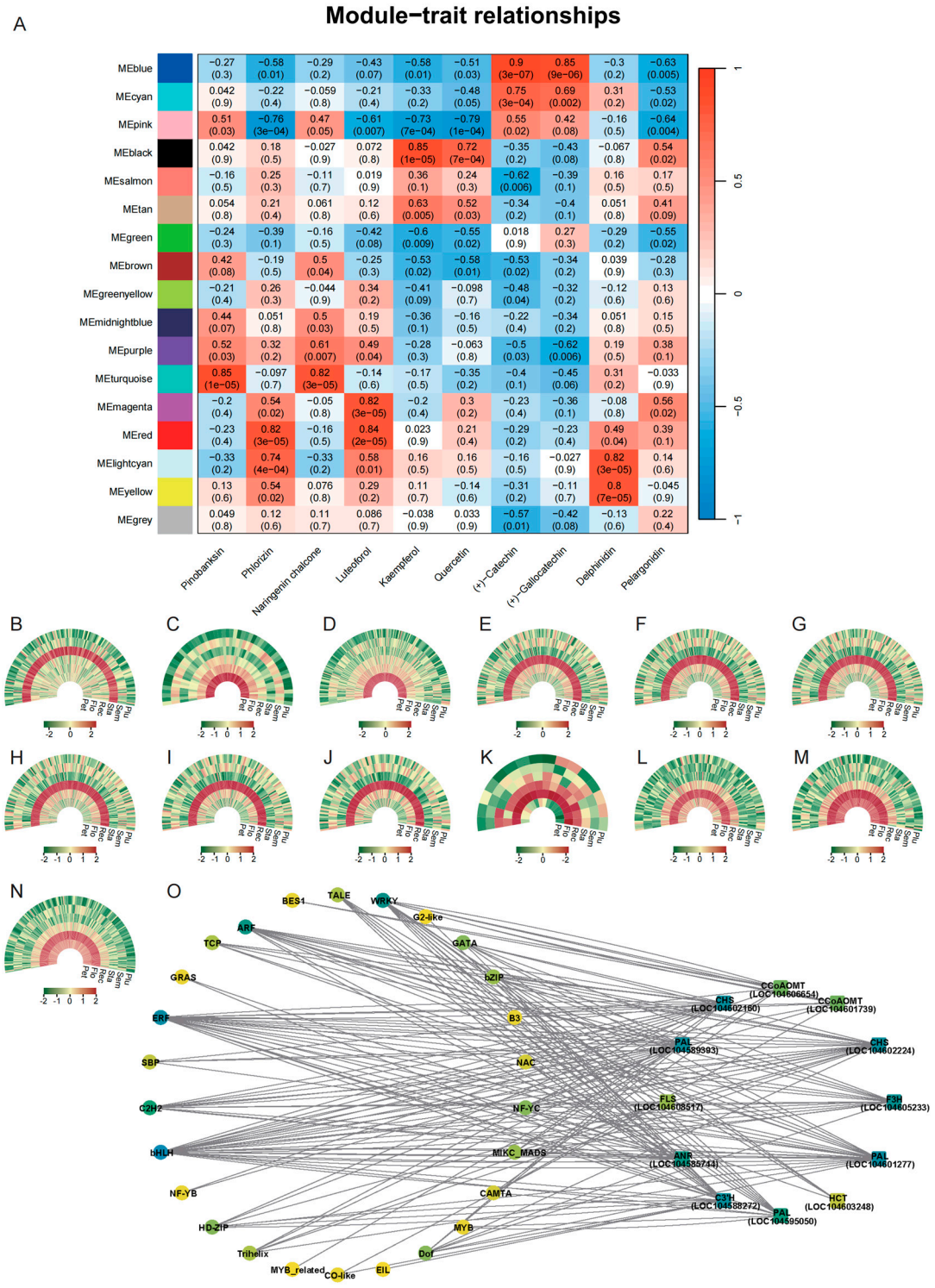


Figure 7. Identification of genes related in flavonoid synthesis of 6 lotus tissues. (**A**) Heatmap of module–trait correlation between major flavonoid components and the module characteristic genes. (**B**) Expression of the structural gene—LOC104608517 (*FLS*) and its interacting genes in module MEblack. (**C**,**D**) Expression of the structural genes—LOC104603248 (*HCT*) and LOC104588272 (*C3′H*) and their interacting genes in module MEturquoise. (**E**–**J**) Expression of the structural genes—LOC104601277 (*PAL*), LOC104589393 (*PAL*), LOC104602160 (*CHS*), LOC104602224 (*CHS*), LOC104605233 (*F3H*), and LOC104585744 (*ANR*) and their interacting genes in module MEmagenta. (**K**) Expression of the structural gene—LOC104611136 (*CCoAOMT*) and its interacting genes in module MElightcyan. (**L**–**N**) Expression of the structural genes—LOC104595050 (*PAL*), LOC104601739 (*CCoAOMT*), and LOC104606654 (*CCoAOMT*) and their interacting genes in module MEyellow. (**O**) Network diagram of transcription factors interaction with structural genes.

Table 2. Module identification.

Module	Structural Genes	Interacting Genes Number	Strongly Correlated Flavonoids
MEblack	LOC104608517 (*FLS*)	154	kaempferol
MEturquoise	LOC104603248 (*HCT*)	54	pinobanksin, naringenin chalcone
MEturquoise	LOC104588272 (*C3′H*)	199	pinobanksin, naringenin chalcone
MEmagenta	LOC104601277 (*PAL*)	189	luteoforol
MEmagenta	LOC104589393 (*PAL*)	173	luteoforol
MEmagenta	LOC104602160 (*CHS*)	181	luteoforol
MEmagenta	LOC104602224 (CHS)	178	luteoforol
MEmagenta	LOC104605233 (*F3H*)	139	luteoforol
MEmagenta	LOC104585744 (*ANR*)	116	luteoforol
MElightcyan	LOC104611136 (*CCoAOMT*)	15	delphinidin
MEyellow	LOC104595050 (*PAL*)	131	delphinidin
MEyellow	LOC104601739 (*CCoAOMT*)	133	delphinidin
MEyellow	LOC104606654 (*CCoAOMT*)	198	delphinidin

Table 3. The top 10 nodes calculated based on the MCC algorithm.

Structural Genes	Interacting Transcription Factors	Strongly Correlated Flavonoids
LOC104601277 (*PAL*)	bHLH,ERF,NF-YC	luteoforol
LOC104602224 (*CHS*)	bHLH,ERF,NF-YC	luteoforol
LOC104589393 (*PAL*)	bHLH,ERF,NF-YC	luteoforol
LOC104588272 (*C3′H*)	bHLH,ERF,NF-YC	pinobanksin, naringenin chalcone
LOC104585744 (*ANR*)	bHLH,ERF,NF-YC	luteoforol
LOC104595050 (*PAL*)	bHLH,ERF	delphinidin
LOC104608517 (*FLS*)	bHLH	kaempferol

4. Discussion

4.1. Relationship Between Tissue Specificity and Pharmacological Effects

In the metabolomics analysis, different tissues of lotus exhibited distinct metabolic features. For instance, Sta differed from other tissues significantly. The distinct accumulation patterns of flavonoids, alkaloids, and organooxygen compounds in various parts of the lotus not only contributed to their unique pharmacological properties but also provide insights into the underlying the mechanism of "homologous and different effects".

Sta has peculiar metabolic features, in which the relative contents of secondary metabolites including amygdalin, liquidambaric, echinatin, and geranial are much higher than that in other tissues. Therefore, this may be the source of its pharmacological antioxidant, anticancer, anti-inflammatory, analgesic, hypoglycemic, and immunomodulatory properties [22,38,47–54]. These metabolites can be considered as key indicators for quality evaluation when developing products related to Sta. Additionally, Sta are enriched with many prenol lipids, which can improve their resistance to salt stress [55].

As light-exposed tissues, Pet, Rec, and Flo have similar metabolic characteristics, which are majorly characterized by the abundant accumulation of flavonoid components. Flavonoids are an important class of antioxidants and their synthesis in plants is strongly influenced by light [56]. Pinocembrin, liquiritigenin, pinobanksin, and naringenin chalcone exhibit various bioactivities, including anti-inflammatory, antimicrobial, anticarcinogenic, antioxidant, neuroprotective, intestinal flora regulatory, antidiabetic, anti hyperuricemic, and antiallergic effects [57–63]. The present study revealed that these compounds were more enriched in Pet than in other tissues, which is important for further comprehensively developing and utilizing Pet as well as for developing quality standards for the traditional Chinese medicine Nelumbinis Petiolus. Dihydroquercetin possess various biological properties, such as antioxidant, antimicrobial, anticancer, anti-Alzheimer's disease, cardioprotective, and hepatoprotective effects [64,65]. Additionally, dihydroquercetin shows promising efficacy in the prevention and treatment of skin-related diseases, such as preventing skin cancer [66,67], treating psoriasis, alleviating skin inflammation [68–71], and delaying skin aging [71,72]. As a natural botanical rich in dihydroquercetin, Rec can be efficiently used in manufacturing skin wound dressings [73]. Delphinidin exerts good pharmacological activities including anticancer, antidiabetic, anti-obesity, cardiovascular protective, and neuroprotective effects, as well as inhibition of abnormal angiogenesis [74,75], which is significantly enriched in Flo. It also affects the antidiabetic and anti-obesity pharmacological effects of Flo.

Although Sem and Plu originate from the same organ with some similarity in metabolite composition, their gene expression patterns are significantly distinct. This shows how small genetic differences can cause significant functional differentiation even in similar general environments. The enrichment of metabolites including terpenoids and proanthocyanidins in Plu may be responsible for its antioxidant capacity [18,76]. In addition, terpenoids enriched in Plu slow down aging and treat senile diseases [77–79], which is the material basis for the anti-aging health effects of Plu. Sem is rich in organooxygen compounds including sugars and lipids, which can provide the body with the required energy and nutrients and boost the body's immunity [11], making it an ideal nourishing ingredient. Sem and Plu are rich in alkaloids, which may be responsible for clearing heat and fire, strengthening the heart, and tranquilizing the mind, as well as exhibiting sedating and hypnotic effects [80–83].

4.2. Tissue-Specific Regulation of Flavonoid Biosynthetic Pathway

Combined metabolomic and transcriptomic analyses revealed a strong connection between metabolite accumulation and gene expression. Specifically, the flavonoid biosynthesis pathway was significantly enriched in several contrasting combinations of tissues, involving 34 differential flavonoid components and 64 key-acting genes; this suggests that this process is under strict genetic control. The high expression of the *CHS*s in Rec may be the mechanism for specific enrichment of flavonoid components in Cluster 9 in this tissue. The high expression of *CHI*s in Pet may contribute to enhanced production of flavonoids such as pinocembrin and liquiritigenin in this tissue. The relative expression of *F3′H*s and *F3′5′H*s in Rec, Sta, and Sem was consistent with the distribution of the content of

dihydroquercetin and dihydromyricetin. We further confirmed the critical role of these genes in flavonoid synthesis. These findings indicate the tissue-specific regulation of the flavonoid biosynthesis pathway and provide a molecular-level mechanism for elucidating differences in the medicinal effects of different parts of lotus.

4.3. Regulatory Role of Transcription Factors in Flavonoid Synthesis

The WGCNA analysis led to the identification of several co-expression modules associated with the flavonoid synthesis, which contained key structural genes and transcription factors involved in the regulation of flavonoid synthesis. Specifically, 26 transcription factor family members, including bHLH, ERF, and NF-YC, were found to interact with several structural genes and participate in the regulation of the synthesis of several flavonoids, including kaempferol, pinobanksin, luteoforol, and delphinidin (Table S4). MYB and bHLH have been implicated in the biosynthesis of flavonoids in Arabidopsis thaliana, maize, tomato, Sechium edulel, and ginkgo [84–90], and can synergistically interact with each other to form an MBW complex with WD40 [85,90,91]. GbMYBR1 modulates flavonoid synthesis in ginkgo. However, its overexpression in Arabidopsis thaliana leads to a reduction in the levels of anthocyanins and flavonols [88]. These previous reports indicate that transcription factors of flavonoid synthesis are modulated by mutual factors and may have different effects in different plants. In addition, the results indicate that transcription factor families such as WRKY, ERF, HD-Zip, ARF, C2H2, bZIP, and NF-YC regulate flavonoid synthesis [90,92].

In this study, two MYB transcription factors were identified in the yellow module which interacted with the *PAL*s to promote the accumulation of delphinidin in Flo. Eight bHLH transcription factors were detected in the black, turquoise, magenta, and yellow modules, interacting with the *FLS*s, *C3′H*s, *PAL*s, *CHS*s, *F3H*s, and *ANR*s and contributing to the positive regulation of kaempferol in Sta, pinobanksin and naringenin chalcone in Pet, luteoforol in Rec, and delphinidin in Flo. Seven ERF transcription factors were identified in the turquoise, magenta, and yellow modules, all of which interacted with the *C3′H*s, *PAL*s, *CHS*s, *F3H*s, *ANR*s, and *CCoAOMT*s, enhancing the accumulation of pinobanksin and naringenin chalcone in Pet, luteoforol in Rec, and delphinidin in Flo. Further analysis revealed that two NF-YC transcription factors were identified in the turquoise and magenta modules, with reciprocal relationships with the *C3′H*s, *PAL*s, *CHS*s, *F3H*s, *ANR*s, which positively regulated pinobanksin and naringenin chalcone in Pet and luteoforol in Rec. Based on these findings, further functional validation and mechanistic studies are recommended to validate the present results and identify potential regulators which could not be identified via the single-omics approach. This will elucidate the mechanism of the regulatory network of lotus metabolism.

4.4. Future Research Directions

In the future, researchers need to explore the synthetic pathways of other metabolites and their regulatory mechanisms to uncover the complexity and dynamics of the metabolic regulatory network of the lotus plant and provide a solid scientific basis for guiding the molecular breeding and rational utilization of lotus. In addition, although some transcription factors associated with flavonoid synthesis have been identified, their specific regulatory mechanisms still need to be further verified. The specific effects of transcription factors on flavonoid synthesis can be elucidated through CRISPR/Cas9 gene editing technology, either by knocking out or overexpressing these factors. In addition, the effects of environmental factors such as light, temperature, and water on the accumulation of lotus metabolites also need to be studied to demonstrate how external conditions influence the metabolic regulatory network of the lotus plant. Lotus tissues may display distinct

developmental patterns, and thus, future studies should conduct time-series analyses to examine metabolite distribution and gene expression across different developmental stages to generate ideas for optimizing the management of lotus throughout its life cycle.

5. Conclusions

In this study, we systematically analyzed metabolite distribution and gene expression profiles of six medicinal parts of the lotus plant and identified the functional genes involved in flavonoid synthesis. Our results provide molecular evidence to support the phenomenon of "homologous and different effects" in lotus plants. Investigating the regulatory mechanisms of the flavonoid biosynthesis pathway and constructing the transcriptional regulatory network of flavonoids may offer new insights into the molecular breeding of lotus plants enriched in specific metabolites, as well as discover novel medicinal benefits of different parts of the lotus plant. Additionally, this work will provide a theoretical basis for the quality control of medicinal materials to meet market demands.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app15052319/s1, Table S1: Quantification table of metabolites identified by MS/MS secondary analysis; Table S2: K-means cluster analysis results; Table S3: Information on structural and interacting genes in key modules; Table S4: Information on structural genes and co-expressed transcription factors in key modules.

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